SIMATS ENGINEERING

ASSESSMENT TOOL-1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

Code of Conduct

I D.TEJA VENKATA SWAROOP(192565098) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : TEJA.D

Date : 06/102026

Analytical Problem Solving

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1. Water Jug Problem

Problem:

You are given two jugs, a 4-gallon jug and a 3-gallon jug. Neither has any measuring marker. A pump can be used to fill the jugs. The objective is to obtain exactly 2 gallons of water in the 4-gallon jug.

Assumptions:

- A jug can be completely filled from the pump.
- Water can be poured from one jug into the other.
- Water can be emptied onto the ground.
- There are no other measuring devices.

State representation: (x, y), where x is the amount of water in the 4-gallon jug and y is the amount in the 3-gallon jug.

Initial state: (0, 0)

Goal state: (2, y), where y can be any amount from 0 to 3 gallons.

Solution steps:

- (0, 0) → (0, 3): Fill the 3-gallon jug.
- (0, 3) → (3, 0): Pour the 3-gallon jug into the 4-gallon jug.
- (3, 0) → (3, 3): Fill the 3-gallon jug again.
- (3, 3) → (4, 2): Pour from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. The 3-gallon jug has 2 gallons left.
- (4, 2) → (0, 2): Empty the 4-gallon jug.
- (0, 2) → (2, 0): Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug.

Final state: (2, 0). Therefore, exactly 2 gallons of water are obtained in the 4-gallon jug.

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2. Mars Rover as an Intelligent Agent

i. What are the percepts for this agent?

The Mars Rover receives information about its surroundings through sensors. Its percepts include images of the terrain, rocks and soil, distance to obstacles, slope and surface conditions, temperature, atmospheric conditions, location, battery level, and scientific measurements from collected samples.

ii. Characterize the operating environment.

- Partially observable – the rover cannot directly observe the entire Martian environment.
- Single-agent – the rover normally operates as the main decision-making agent in its local task.
- Sequential – current actions affect future positions, energy and scientific results.
- Dynamic – environmental conditions can change while the rover is operating.
- Continuous – position, movement, time and many physical measurements vary continuously.
- Mostly deterministic for planned actions, but uncertainty exists because terrain and equipment conditions are not perfectly known.
- Unknown/partially known – the rover must explore and learn about its surroundings.

iii. What are the actions the agent can take?

The rover can move forward or backward, turn, stop, navigate around obstacles, capture images, collect soil or rock samples, analyse samples using scientific instruments, transmit data to Earth, monitor its systems, and conserve or recharge energy where possible.

iv. How can one evaluate the performance of the agent?

Performance can be evaluated using a performance measure such as the amount and quality of scientific data collected, number of useful samples analysed, distance successfully explored, accuracy of navigation, avoidance of obstacles, energy efficiency, communication success, equipment safety, and completion of mission objectives.

v. What sort of agent architecture is most suitable? Why?

A model-based, goal-based and utility-based agent with a learning component is most suitable. The rover must maintain an internal model because it cannot observe everything directly. It must use goals such as reaching a location or analysing a sample, while a utility function can help it choose safer and more scientifically valuable actions. Learning allows it to improve its decisions from experience.

3. 8-Queens Problem

Problem formulation:

The 8-Queens problem requires placing 8 queens on an 8 × 8 chessboard so that no two queens attack each other. Therefore, no two queens can be in the same row, same column, or on the same diagonal.

Problem components:

- Initial state: An empty 8 × 8 chessboard.
- State: A partial or complete arrangement of queens on the board.
- Actions/operators: Place a queen in a row/column position that does not conflict with already placed queens.
- Transition model: Adding a non-attacking queen creates a new state.
- Goal test: Exactly 8 queens are placed and no pair of queens attacks each other.
- Path cost: Usually each queen placement has unit cost, so a solution with 8 placements has cost 8.

Search strategies:

- Breadth-First Search (BFS): Explores states level by level. It can find a solution but uses large memory.
- Depth-First Search (DFS): Places queens one by one and backtracks when a conflict occurs. It is simple and memory efficient.
- Backtracking search: A depth-first approach that immediately rejects a partial arrangement when a conflict is detected.
- Heuristic search: Can use measures such as the number of attacking queen pairs to guide the search toward a solution.

A practical approach is backtracking search because it reduces the search space by rejecting invalid partial configurations as early as possible.

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4. OLA Cab Booking – Problem-Solving Agent

Problem:

A customer wants to travel from one location to another using an OLA Cab booking mobile application. The customer can choose among mini, micro, sedan, shared, prime and other available cab types according to comfort and destination requirements.

Type of problem-solving agent:

A utility-based goal-oriented agent is most suitable. The agent has the goal of taking the customer to the destination, while utility can be used to compare fare, travel time, comfort, availability and other customer preferences

Pseudocode:

START

Receive source location and destination.

Receive customer preferences: cab type, comfort, budget and maximum acceptable travel time.

Get available cabs and their estimated fares and travel times.

FOR each available cab:

  Check whether it satisfies the customer's constraints.

  Calculate a utility score using fare, time, comfort and preference.

END FOR

Select the cab with the highest utility score.

Display cab type, estimated fare, driver/cab details and estimated arrival time.

IF customer confirms booking:

  Book the selected cab.

  Track the cab and update the customer.

ELSE:

Allow the customer to modify preferences and repeat selection.

END

5. Uniform Cost Search (UCS) – Logistics Delivery Network

Problem:

A logistics company wants to find the least-cost path from starting warehouse S to destination warehouse G. The graph is weighted, and each edge represents travel cost.

Edges from the given graph:

- S → A = 1
- A → B = 3
- A → C = 1
- B → D = 3
- C → D = 1
- D → G = 3
- C → G = 2
- S → G = 12

Uniform Cost Search principle:

UCS always expands the node with the smallest accumulated path cost g(n). It continues until the goal node is selected for expansion, guaranteeing the least-cost solution when all edge costs are non-negative.

UCS expansion:

- Start: S, cost = 0.
- Expand S → A (cost 1), G (cost 12). Frontier: A(1), G(12).
- Expand A(1) → C(2), B(4). Frontier: C(2), B(4), G(12).
- Expand C(2) → G(4), D(3). Frontier: D(3), B(4), G(4).
- Expand D(3) → G(6). Existing G(4) is cheaper, so keep G(4).
- The next lowest-cost node is G(4), so the goal is reached.

Possible path costs:

- S → G = 12
- S → A → B → D → G = 1 + 3 + 3 + 3 = 10
- S → A → C → D → G = 1 + 1 + 1 + 3 = 6
- S → A → C → G = 1 + 1 + 2 = 4

Optimal path: S → A → C → G

Minimum total cost = 4.

Interpretation of result:

Using Uniform Cost Search, the least-cost route from warehouse S to warehouse G is S → A → C → G with a total travel cost of 4 units. This is better than the direct route S → G, which costs 12 units, and all other available routes shown in the graph.
